

DDQN-Enabled UAV Path Planning in Non-Terrestrial Networks Under ICAS Constraints

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Abstract—Non-Terrestrial Networks (NTNs) are aimed to integrate satellites, high-altitude platforms, and UAVs for seamless global connectivity towards 6G wireless. However, optimal UAV trajectory within multi-layer NTN environments remains challenging due to dynamic user distribution, link variability, and strict energy constraints. In this work, we propose a Dueling Double Deep Q-Network (DDQN)-based reinforcement learning framework that jointly optimizes access and backhaul link quality, user coverage, and energy efficiency under Integrated Communication and Sensing (ICAS) constraints. A custom 3D simulation environment with channel-aware signal potential (SP) maps and multi-objective reward formulation is developed to emulate the realistic NTN dynamics. The proposed dueling DDQN achieves up to 90% mission success rate, faster convergence, and substantially reduced collision frequency as compared to other reinforcement learning approaches, highlighting its suitability for reliable and energy-efficient UAV navigation in dynamic NTN scenarios.

Index Terms—UAV trajectory optimization, deep reinforcement learning, dueling double DQN, non-terrestrial networks, integrated communication and sensing.

I. INTRODUCTION

With the advancement of 6G, Non-Terrestrial Networks (NTNs) have emerged as a key enabler for achieving seamless global connectivity with enhanced network coverage [1], [2]. Unlike conventional terrestrial networks, NTNs integrate satellites, high-altitude platforms (HAPs), and unmanned aerial vehicles (UAVs) into a unified architecture, thereby extending coverage to remote, disaster-affected, or infrastructure-deficient areas [1], [3]. In such multi-layered environments, UAVs serve as agile low-altitude aerial relays capable of dynamically adapting their positions to ensure seamless access and backhaul communication between ground users and satellite segments [4], [5].

However, it is difficult to achieve optimal UAV trajectory planning in NTN scenarios, with simultaneous maximum user coverage, maintaining dual-link reliability (UAV-satellite and UAV-ground), avoiding obstacles, and conserving energy—all under dynamic channel and environmental conditions [6]. Traditional optimization techniques often fail to adapt in such settings due to the stochastic nature of wireless channel links and the high dimensionality of the search space. This calls for the use of learning-driven strategies that can autonomously optimize UAV movement in complex NTN conditions [7].

A. State of the Art

Conventional UAV path planning approaches, including A* search, Dijkstra's algorithm, and heuristic optimization, are although computationally efficient but they are inherently static. Hence, they are less effective in dynamic 3D NTN environments [5], [8]. In contrast, reinforcement learning (RL)-based approaches are prominent in handling such high-dimensional problems by enabling UAVs to learn optimal movement policies through interaction with the environment [4], [7]. Among RL algorithms, Q-learning and Deep Q-Networks (DQN) have been widely explored for UAV trajectory optimization and resource allocation tasks [9], [10]. However, standard DQNs suffer from instability and overestimation of Q-values during training, limiting their scalability in complex NTN scenarios. To address these issues, Double DQN (DDQN) and Dueling DQN architectures have been introduced, which improve learning stability and decision accuracy by decoupling value estimation and policy evaluation [11], [12]. But their integration into NTN-based UAV trajectory optimization remains underexplored.

B. Research Gap and Motivation

Existing RL-based UAV path planning frameworks primarily focus on terrestrial or single-layer aerial networks and often overlook the dual-link dynamics and Integrated Communication and Sensing (ICAS) constraints inherent in NTNs [3]. The absence of coordinated optimization between the UAV-ground access link and the UAV-satellite backhaul link leads to suboptimal performance in terms of coverage, reliability, and energy efficiency. To address this gap, there is a need for intelligent multi-objective learning framework which is capable of optimizing UAV trajectories under the realistic NTN constraints, where decisions must jointly consider link quality, energy, and spatial exploration.

C. Novelty and Scope

In this paper, we propose a Dueling Double Deep Q-Network (DDQN) based UAV trajectory optimization framework designed for multi-layer NTN environments. The proposed approach integrates aerial path learning with a multi-objective reward function that balances communication reliability (both access and backhaul), obstacle avoidance, and energy conservation. To our best knowledge, this is the first

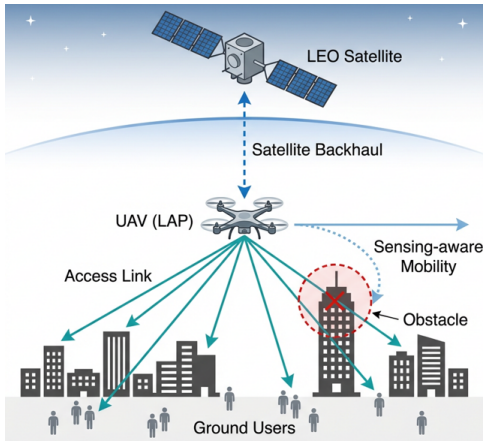


Fig. 1. System model

work which employ a DDQN-based agent in a 3D NTN-inspired environment that reflects realistic link conditions via dynamic signal potential (SP) maps. The proposed framework provides an adaptive and scalable solution for autonomous UAV operation, forming a key step towards AI-driven NTN deployment for beyond-5G and 6G systems.

D. Major Contributions of the Work

The key contributions of this work are:

- 1) A dueling DDQN-based reinforcement learning framework is proposed for UAV trajectory optimization tailored to multi-layer NTN conditions.
- 2) A multi-objective reward function is formulated which jointly optimizes the link quality, user coverage, collision avoidance, and energy efficiency.
- 3) A custom NTN-inspired 3D simulation environment integrating user distribution, obstacle maps, and dual-link SP dynamics for realistic training and evaluation.
- 4) Comprehensive comparative analysis with baseline DQN and DDQN methods, demonstrating improved mission success, stability, and safety.

II. SYSTEM MODEL

We consider a multi-layer NTN comprising a Low Earth Orbit (LEO) satellite, a single unmanned aerial vehicle (UAV), and terrestrial user regions. The UAV operates as a low-altitude aerial platform (LAP), simultaneously maintaining satellite backhaul connectivity and offering access coverage to ground users, as shown in Fig. 1. Sensing representation is abstracted through obstacle-awareness and blockage detection; UAV's obstacle avoidance plays the role of sensing-aware mobility. Here SNR represents SP only.

The UAV navigates inside a bounded three-dimensional region of size $20 \times 20 \times 5$, representing an NTN-assisted urban subspace. Each coordinate (x, y, z) corresponds to a feasible UAV location, where the ground plane is at $z = 0$ and the UAV flies at $z > 0$. The discretized 3D environment used for simulation is shown in Fig. 2.

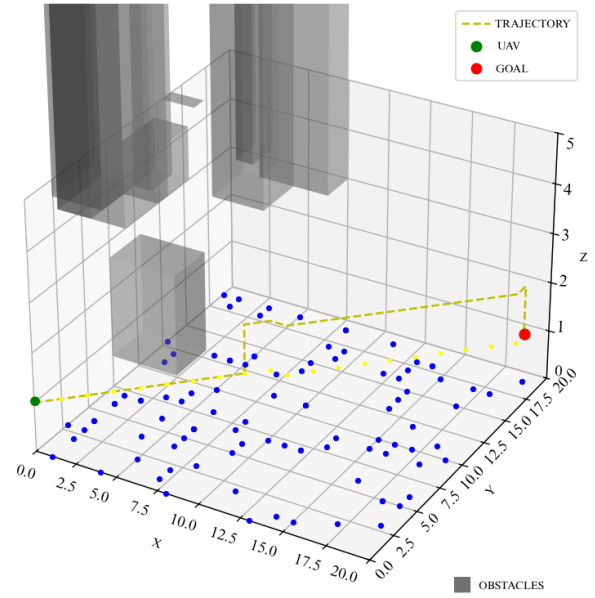


Fig. 2. UAV 3D View

A. Communication Entities

The system consists of three primary components:

- **LEO Satellite (Backhaul Node):** Positioned at altitude H_s , the satellite provides wide-area NTN backhaul illumination. The UAV must maintain sufficient satellite-UAV SNR γ_{su} throughout flight, as depicted in Fig. 1.
- **UAV (Low-Altitude Relay):** UAV operates at altitude H_u , and can move horizontally across the grid. It forms an intermediate link between NTN backhaul and ground access.
- **Ground User Regions:** User clusters are distributed on ground plane ($z = 0$) and abstracted as several signal sources placed randomly within the grid in each simulation run. These sources represent localized access-demand hotspots.

B. Unified SP Field Representation

Instead of separately modeling LEO-UAV and UAV-ground link budgets, we adopt a unified SP field over the 3D grid, capturing the combined effect of the satellite backhaul illumination (elevated, wide-area component), and ground-user clusters (localized access components).

A small number of signal sources, placed randomly at each experiment, generate a spatial SP field $\mathcal{S}(x, y, z)$ that serves as a proxy for combined NTN backhaul illumination and access hotspots:

$$\mathcal{S}(x, y, z) = \sum_{k=1}^{N_s} \frac{P_k}{\|(x, y, z) - (x_k, y_k, z_k)\|^\alpha + \epsilon}, \quad (1)$$

where N_s is the number of sources, (x_k, y_k, z_k) their coordinates, P_k their transmit powers, α the path-loss exponent,

and ϵ a small regularization term. In Fig. 1, the high-level NTN interpretation of this field is shown through backhaul ($\mathcal{S}_{su}$) and access ($\mathcal{S}_{ug}$) links. It is important to note that (1) defines a spatial SP field used to emulate heterogeneous NTN connectivity conditions for reinforcement learning. This abstraction allows the UAV to learn navigation policies without relying on specific channel models, while still preserving realistic spatial variability of communication quality. In Fig. 2, its discretized form is visualized for simulation.

C. Obstacle and Blockage Model

Urban obstacles such as buildings are represented using a binary occupancy grid. Each occupied cell denotes a no-fly zone for UAV navigation, and a radio obstruction, reducing SP due to NLoS conditions. These structures appear as shaded blocks in both figures.

D. UAV State and Action Space

At time step t , the UAV state is defined as:

$$s_t = (x_t, y_t, z_t, \mathcal{S}(x_t, y_t, z_t), \text{ObstacleMap}), \quad (2)$$

capturing the UAV's position, corresponding channel quality, and surrounding environment.

UAV can hover or move using a discrete action set:

$$\mathcal{A} = \{\text{North, South, East, West, Up, Down, NE, NW}\}.$$

Only transitions to valid, non-obstacle cells are allowed.

E. NTN-Aware Objective

The UAV aims to learn a trajectory that satisfies the following joint objectives:

- 1) Maintain NTN backhaul availability (high γ_{su}).
- 2) Maximize access connectivity to ground-user regions (high γ_{ug}).
- 3) Avoid obstacles and restricted regions.
- 4) Minimize unnecessary movement to conserve energy.

These combined NTN-aware criteria form the basis for the DDQN-based trajectory optimization formulation. The corresponding reward can be mathematically expressed as,

$$r_t = w_1 R_\gamma(t) + w_2 R_b(t) + w_3 R_u(t) + w_4 R_o(t) + w_5 R_m(t), \quad (3)$$

where $R_\gamma(t)$ denotes unified SP-based reward, $R_b(t)$ is backhaul (satellite link) reward, $R_u(t)$ represents access/coverage reward for ground users, $R_o(t)$ is obstacle penalty, $R_m(t)$ is movement/energy penalty, and $w_i \forall i \in [1, 5]$ are the weights controlling the contribution of each component. This structure allows the UAV to balance communication quality, sensing-aware safety, and motion efficiency.

III. PROPOSED METHOD: DDQN-BASED NTN-AWARE UAV TRAJECTORY OPTIMIZATION

Based on the NTN system model described in Section II and the environment wrapper described in Algorithm 1, the UAV learns a navigation policy that maximizes access and backhaul connectivity while avoiding obstacles and minimizing energy

expenditure. The reinforcement learning agent interacts with the environment through the UAVenv wrapper (see Algorithm 1) which implements state transitions, SP lookups, reward computation and termination checks. The learning algorithm is a Dueling DDQN tailored for the discrete grid action space defined in Section II-D.

A. MDP and Environment Interaction

The MDP remains $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, r, \gamma)$ with states and actions defined in Section II-D. At each time step t , the agent:

- 1) observes current state s_t ,
- 2) selects action a_t using an ϵ -greedy policy derived from the online Q-network,
- 3) calls the environment wrapper: $(s_{t+1}, r_t, done) \leftarrow \text{UAVenv}(s_t, a_t)$,
- 4) stores transition $(s_t, a_t, r_t, s_{t+1}, done)$ into the replay buffer.

Using Algorithm 1 as the canonical environment interface ensures that SP evaluations (both backhaul and access components), obstacle checks, and termination logic are consistently applied during both training and evaluation.

B. Network Architecture and Q-value Estimation

We use a dueling Q-network architecture with parameters θ (online) and a target network with parameters θ^- . The network processes the state vector s_t (position indices, local SP value and a small local obstacle patch) and outputs Q-values for all actions $a \in \mathcal{A}$. The dueling head separates the state-value and advantage streams as,

$$Q_\theta(s, a) = V_\theta(s) + \left(A_\theta(s, a) - \frac{1}{|\mathcal{A}|} \sum_{a'} A_\theta(s, a') \right). \quad (4)$$

where $Q_\theta(s, a)$ denotes the state-action value estimated by the dueling network with parameters θ , $V_\theta(s)$ represents the state-value function, capturing the overall utility of being in state s . The advantage function $A_\theta(s, a)$ quantifies the relative benefit of taking action a in state s , and the mean subtraction term $\frac{1}{|\mathcal{A}|} \sum_{a' \in \mathcal{A}} A_\theta(s, a')$ ensures identifiability by normalizing the advantages across all actions. Finally, $\mathcal{A}$ is the discrete action set available to the UAV, and $|\mathcal{A}|$ is its cardinality.

C. DDQN Target and Loss

To reduce overestimation, we adopt the DDQN target. Given a sampled mini-batch of transitions $(s_j, a_j, r_j, s'_j, done_j)$ from the replay buffer, the target for each sample is computed as:

$$y_j = r_j + (1 - done_j) \gamma Q_{\theta^-}(s'_j, \arg \max_a Q_\theta(s'_j, a)). \quad (5)$$

The online network is updated by minimizing the mean squared TD error:

$$L(\theta) = \frac{1}{B} \sum_{j=1}^B (y_j - Q_\theta(s_j, a_j))^2, \quad (6)$$

where B denotes the mini-batch size.

D. Training Procedure

The DDQN agent interacts with the environment exclusively through the UAVenv function (Algorithm 1), which provides state transitions, unified NTN-aware SP values, reward calculation and episode termination. The agent maintains an online Q-network Q_θ and a target network Q_{θ^-} along with a replay buffer for stabilizing training.

At the beginning of each episode, the UAV is placed at a random valid grid cell. At every time step, the agent selects an action using an ϵ -greedy policy based on Q_θ . The environment is updated as,

$$(s_{t+1}, r_t, \text{done}) \leftarrow \text{UAVenv}(s_t, a_t), \quad (7)$$

and the transition (s_t, a_t, r_t, s_{t+1}) is stored into the replay buffer. Mini-batches sampled from this buffer are used to update the online network using the DDQN target:

$$y_t = r_t + (1 - \text{done})\gamma Q_{\theta^-}(s_{t+1}, \arg \max_a Q_\theta(s_{t+1}, a)). \quad (8)$$

The target network parameters are periodically synchronized with the online network. Exploration is gradually reduced by decaying ϵ over training episodes. Training continues until the cumulative reward and policy behavior converge. Further, experience replay with uniform sampling is used to break correlation between consecutive transitions. Moreover, typical hyperparameters used in our experiments: learning rate $\alpha_{lr} = 0.001$, discount factor $\gamma = 0.99$, mini-batch size $B = 64$, target update period T_{update} , replay capacity $|\mathcal{D}| = 10^5$, and initial/final ϵ values for exploration. The overall computational load is primarily determined by the neural-network training and replay-buffer sampling steps, whereas the inference phase incurs negligible cost, making the learned policy suitable for real-time UAV execution.

E. Integration with UAVenv (Algorithm 1)

Use of UAVenv interface ensures a clear separation between the environment dynamics and the DDQN learning process. The agent receives only the observable state, reward, and termination signal from the environment, without accessing internal details such as the underlying SP-source distribution or obstacle layout. This design guarantees consistent transition and reward semantics throughout training and evaluation, and facilitates controlled experimentation by allowing modifications to specific environment components without altering the learning algorithm. During evaluation, a greedy or near-greedy policy is employed, and performance is assessed using standard metrics, including mission success rate, cumulative reward, collision frequency, and the fraction of time for which the satellite backhaul condition $\gamma_{su} \geq \gamma_{su}^{\min}$ is satisfied.

IV. RESULTS AND DISCUSSION

The performance of the proposed DDQN-based NTN-aware UAV path optimization framework is evaluated using multiple metrics, including reward evolution, collision behavior, and task success rate. The results also compare different deep Q-learning variants (DQN, DDQN, and Dueling DDQN) and examine the influence of the reward-weighting parameter β .

Algorithm 1: UAVenv Environment for NTN-Aware UAV Navigation

Input: $s_t = (x_t, y_t, z_t, \mathcal{S}_t, \text{ObstacleMap})$, action a_t ,
SP map, obstacle grid

Output: $s_{t+1}, r_t, \text{done}$

done $\leftarrow$ False;

$(x', y', z') \leftarrow f_{\text{move}}(x_t, y_t, z_t, a_t)$;

if cell (x', y', z') is an obstacle **then**

$r_t \leftarrow r_{\text{obstacle}}$;

done $\leftarrow$ True;

return (s_t, r_t, done) ;

$S' \leftarrow \mathcal{S}(x', y', z')$;

Compute γ_{su} and γ_{ug} implied by S' ;

Compute reward components:

$r_S, r_{\text{backhaul}}, r_{\text{access}}, r_{\text{movement}}$;

$r_t \leftarrow r_S + r_{\text{backhaul}} + r_{\text{access}} + r_{\text{movement}}$;

if UAV reaches target region or step limit reached **then**

done $\leftarrow$ True;

$s_{t+1} \leftarrow (x', y', z', S', \text{ObstacleMap})$;

return $(s_{t+1}, r_t, \text{done})$;

All simulations are conducted over 3000 episodes in the grid-based environment introduced earlier.

Fig. 3 presents an example UAV trajectory in the obstacle-rich grid environment. The UAV begins at the lower-left corner and must reach the goal located at the upper right, while navigating through several grey obstacle blocks. The learned trajectory shows that the UAV quickly develops a preference for safer and more structured movement patterns, avoiding dense obstacle clusters and selecting smoother routes toward the target. The scattered blue points represent the underlying SP sources, and the trajectory progression indicates that the agent implicitly leverages higher-SP corridors while maintaining clearance from obstacles. Although occasional collisions occur (as highlighted by the red marker), the agent typically adapts and recovers as training proceeds, demonstrating its ability to trade off path length, SP advantage, and obstacle avoidance in an NTN-aware environment.

Training performance across different algorithms can be observed in Fig. 4, which shows the cumulative rewards achieved by DQN, DDQN, and Dueling DDQN. All three methods exhibit clear upward trends, but the standard DQN progresses slowly and displays higher oscillations, reflecting its known Q-value overestimation issues. DDQN significantly stabilizes the learning curve by decoupling action selection and evaluation, resulting in smoother improvement throughout training. The Dueling DDQN achieves the highest overall reward accumulation and maintains the most consistent high-level plateau, which indicates that the dueling architecture is particularly effective in evaluating state values in the NTN-aware environment. These observations confirm that, with respect to reward performance, the Dueling DDQN provides the strongest learning capability among the evaluated methods.

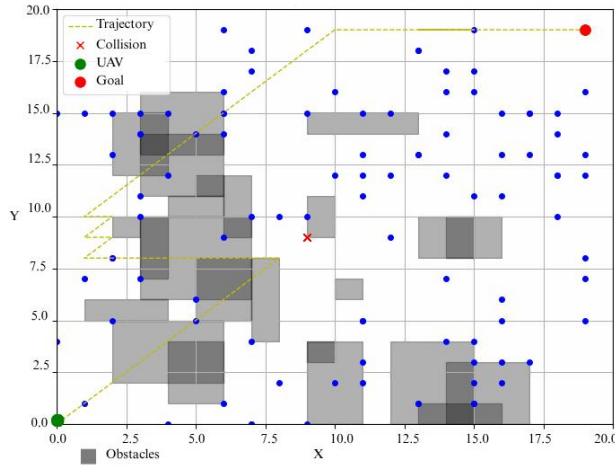


Fig. 3. UAV trajectory visualization for the Dueling DDQN agent ($\beta = 4$).

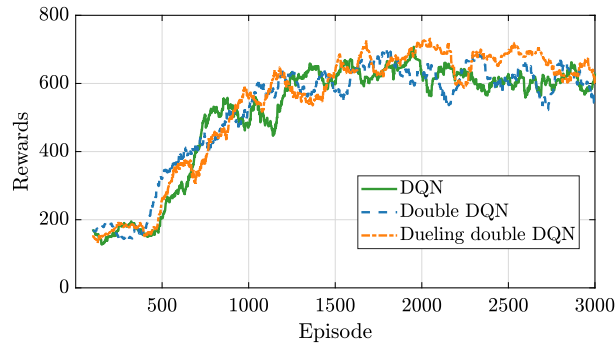


Fig. 4. Training rewards for DQN, DDQN, and Dueling DDQN.

Collision behavior is presented in Fig. 5. Although all algorithms start with relatively high collision frequencies due to unstructured exploration, their long-term behaviors diverge meaningfully. DQN maintains comparatively higher and more erratic collision counts throughout training, showing difficulty in consistently avoiding obstacles. DDQN achieves better stability, reducing collision spikes but still experiencing intermittent peaks. The Dueling DDQN demonstrates the lowest sustained collision levels, especially beyond the mid-training phase, often operating at only one or two collisions per episode. This reflects its superior ability to infer safer paths through obstacle-dense regions, aided by its enhanced state-value decomposition. Thus, in terms of collision reduction and safety, Dueling DDQN provides the best performance.

The effect of β on collision behavior is illustrated in Fig. 6. When $\beta = 0$, the UAV receives no penalty for excessive movement and thus explores more aggressively, leading to higher collision levels. Increasing β to moderate values such as 4 or 6 produces more balanced behavior, with substantially fewer collisions as the UAV becomes more disciplined in selecting lower-risk directions. However, higher $\beta = 10$ imposes excessive penalties on movement, causing the agent to favor shorter paths that occasionally cut too close to obstacles, which

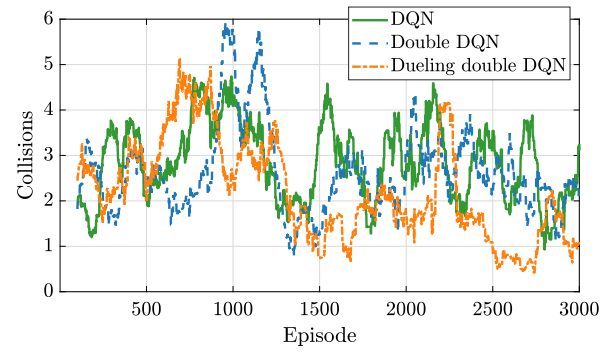


Fig. 5. Collision trends for DQN, DDQN, and Dueling DDQN during training.

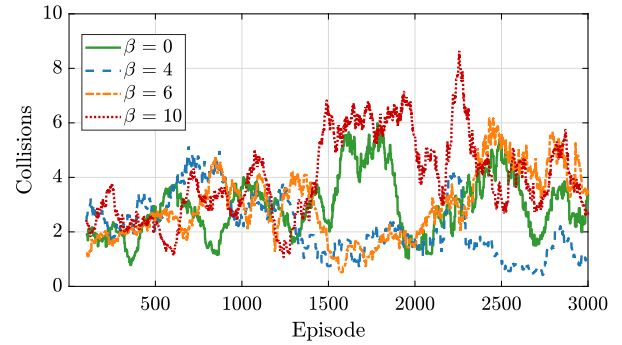


Fig. 6. Collision trends across different β values during training.

increases collision peaks during several phases of training.

The impact of β on cumulative reward is examined in Fig. 7. When $\beta = 0$, the reward trajectory exhibits slow and unstable growth due to unstructured exploration. As β increases to 4 and 6, the agent begins to demonstrate stronger and more consistent improvement, ultimately achieving significantly higher reward values compared to the zero-penalty case. Interestingly, $\beta = 10$ yields the highest cumulative reward overall, particularly after the first 1000 episodes, as the UAV is strongly encouraged to follow energy-efficient and SP-favorable paths. However, the earlier collision plot indicates that excessively large β may lead to riskier shortcuts. Hence, while larger β boosts reward accumulation, a moderate value yields the best balance between safety and efficiency.

Task success rates for different algorithms are shown in Fig. 8. While DQN is slow to achieve reliable task completion and exhibits noticeable instability in later episodes, both DDQN and Dueling DDQN converge rapidly toward high success rates. DDQN demonstrates particularly fast early improvement, reaching success rates above 0.9 earlier than the other methods. The Dueling DDQN follows closely and maintains comparable high performance in the later episodes, with both methods converging to similar final success levels. These results indicate that, although the dueling architecture offers advantages in reward and collision behavior, the overall task-completion capability is shared almost equally between DDQN and Dueling DDQN, both of which significantly outperform

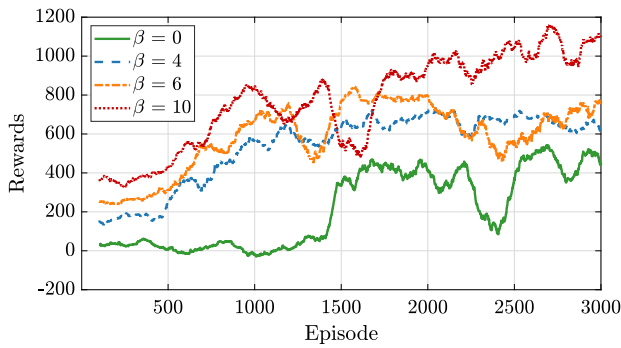


Fig. 7. Reward comparison across different β values during training.

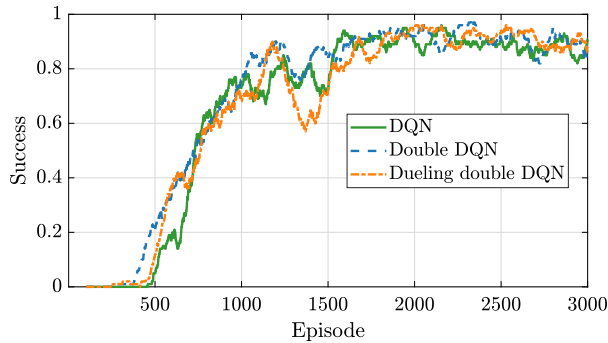


Fig. 8. Success rate comparison for DQN, DDQN, and Dueling DDQN.

the standard DQN baseline.

Finally, Fig. 9 depicts success rate evolution as a function of β . Moderate values ($\beta = 4$ and $\beta = 6$) produce the most satisfactory success trends, with rapid growth and sustained high success levels beyond 0.9. High $\beta = 10$ initially produces fast success gains but also shows occasional dips due to overly constrained movement. For $\beta = 0$, the success rate improves significantly only in the later stages of training, indicating that the absence of movement penalization leads to inefficient exploration in obstacle-rich and SP-varying environments.

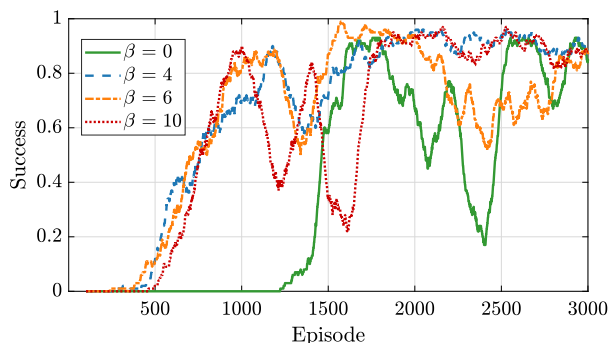


Fig. 9. Success rate trends across different β values during training.

V. CONCLUSION

This work proposed a DDQN-based UAV trajectory optimization framework for NTN under ICAS constraints. By modeling the environment through a unified SP field capturing both satellite backhaul and terrestrial access conditions, with obstacle-aware sensing, UAV learns navigation policies that jointly satisfy communication and sensing requirements. Dueling DDQN variant achieved the highest reward and the lowest collision rates, while both DDQN and Dueling DDQN converged to similar high success rates, demonstrating their effectiveness in complex NTN environments.

In nutshell, the results confirm that deep reinforcement learning is a promising approach for ICAS-consistent UAV navigation in NTNs, enabling safe mobility while maintaining reliable communication performance. Future extensions will explore multi-UAV cooperative strategies, continuous control, and more realistic ICAS constraints involving dynamic sensing demands and interference-aware trajectory planning.

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